

FRANKLINE MISANGO OYOLO

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Education

The University of Hong Kong

Aug 2022 – Jun 2027

BEng. Artificial Intelligence & Data Engineering — GPA: 3.73/4.3

Hong Kong SAR

Professional Experience

The Swiss Federal Institute of Technology - Lausanne

Jul 2025 – August 2025

PhD Research Assistant - Machine Learning and Advanced Analytics

Switzerland

- Developed Diffusion Convolutional RNN (DCRNN) for Swiss bike-share forecasting using EPFL HPC clusters, achieving 27% RMSE reduction vs ST-GCN
- Engineered multi-scale OSM feature integration (500m/1000m/1500m radii) across 1k+ stations — Built 10M+ trip dataset with population grids — Demonstrated 13.4% gain from spatial feature fusion
- Designed attention-based interpretability framework revealing station-type specific patterns (local vs regional focus) — Deployed DCGRU models with Kubernetes/ArgoCD pipelines reducing serving latency by 63%

BASF SE

Jan 2025 – Jun 2025

Business Intelligence Analyst - Global Supply Chain and Digital Business Solutions

Hong Kong

- Architected real-time supply chain dashboard using React/Redux with TensorFlow.js LSTM models (91% forecast accuracy) processing 14TB+ logistics data, reducing stockouts by 38% through automated replenishment signals
- Optimized Spark ETL pipelines using Parquet columnar storage & Airflow dynamic task mapping, achieving 88% memory efficiency (2.4× throughput) on 64-node EMR cluster handling 500GB/day shipments
- Engineered hybrid search system combining Azure Cognitive Search with custom BERT-As-Service (96.2% NDCG@10), reducing NLP query latency 42% via FPGA-accelerated ONNX runtime deployments

Manulife Financial Corporation

Jun 2024 – Aug 2024

Data Governance Summer Analyst

Hong Kong

- Developed enterprise data catalog using React/Node.js with ArangoDB graph backend, achieving 87.2% automated classification of 2.3M+ assets through TF-IDF/Word2Vec similarity scoring at 1.2k QPS
- Redesigned Spring Boot API layer using concurrent Neo4j traversals & Hazelcast caching (40% latency reduction), handling 450 RPS with P99 of less than 120ms on 16-core Azure instances
- Implemented testing framework with Jest/Cypress & Jenkins pipeline integration (98% coverage), catching 142 critical bugs pre-production through mutation testing & chaos engineering scenarios

Department of Architecture, University of Hong Kong

Dec 2023 – Apr 2024

PhD Research Assistant - Machine Learning and Analytics

Hong Kong SAR

- Engineered Meta SAM transformer-based instance segmentation pipeline (ViT-H backbone, 632M parameters) achieving 91.8% IoU on 1000+ architectural floor plans for aeration feature analysis in microbial containment studies
- Implemented distributed coordination algorithm for 200-node swarm robotics system using ROS2 Galactic and ZeroMQ pub/sub architecture, with 5ms latency for real-time airflow monitoring across 100+ housing units
- Fine-tuned DeepLab 3+ with Xception-65 backbone (26.5M parameters) on segmented thermal imagery dataset (n=4,800) for temporal prediction of ventilation efficiency, achieving 0.87 R² on 3-year forecasts
- Architected Kubernetes-orchestrated MLOps pipeline with COCO annotation standard, reducing data preprocessing time by 78% and enabling automated multi-GPU training across 3 distributed DL workloads

TalentLabs Asia Limited

Jun 2023 – Aug 2023

Data Engineer

Hong Kong SAR

- Built Analytics dashboard using LookerML processing 12k+ user events, improving UI conversion by 75% through A/B testing
- Automated GCP ETL pipelines with Prefect orchestrator, reducing weekly reporting time from 20h to 2h
- Developed distributed web scraper harvesting 8k+/day HK job postings via Python Scrapy/Selenium with 2CAPTCHA integration

Projects

NLG Financial Sentiment Analysis | Python, Transformers, LLM Engineering

2024

- Architected multi-model NLP pipeline comparing BERT (84.2% F1), BART (89.1% F1), and Llama-2-70B for financial document analysis, processing 50K+ news articles/day from 37 sources using HuggingFace Transformers and FAISS vector search
- Implemented LoRA fine-tuning with 8-bit quantization (QLoRA), reducing BART model size by 63% while maintaining 98.7% of original accuracy on FiQA benchmark dataset
- Developed custom attention masking for financial jargon handling, improving rare term (frequency <0.1%) sentiment accuracy by 41% compared to baseline BERT architecture

SERSi Spacecraft Algorithm | Python, NASA Data, Planetary Science

2025

- Devised signal reconstruction pipeline leveraging wavelet-based denoising (Daubechies-4) with Maxwell-Boltzmann distribution fitting to recover radiation-corrupted spectroscopic albedo measurements, reducing data loss by 78.3% across 87TB of iPAC/TESS multi-spectral imagery
- Formulated Bayesian uncertainty quantification framework with MCMC sampling (n=10⁷ iterations) for exoplanet habitability indices, achieving AUC=0.94 on ESI prediction across 1,400+ KOI candidates with 27 spectral features

DeepRacer | Python, Reinforcement Learning, ROS2, AWS

2024

- Developed modular reinforcement learning architecture with dynamic action space definitions and reward function templates, achieving 8.4s/lap times through hyperparameter-optimized training (256-episode batches) using AWS SageMaker and ONNX runtime deployment
- Implemented track-specific path planning with Bezier curve optimization, enabling 72km/h straightaways and 45° turn negotiation through progressive steering PID control ($K_p=0.7$, $K_i=0.01$, $K_d=0.03$), ranked top 5% in AWS Global Virtual Circuit among 14K teams
- Designed distributed training system using deepracer-on-the-spot framework with dual-node architecture (Nginx load balancing), processing 18TB training data through automated log analysis pipelines with 92% sample efficiency

BRH35 Brain Cancer Detection | *Flask, CNN, Transfer Learning*

2024

- Constructed distributed Flask microservice architecture (Gunicorn WSGI, 12-worker Nginx load balancing) with MobileNetV3-Large backbone (5.4M parameters) fine-tuned on 3D MRI volumetric data achieving 94.7% classification accuracy with 18ms inference latency
- Integrated HL7-compliant PACS integration layer using DICOM networking protocol (C-STORE SCP) with lossless JPEG2000 compression, enabling 14TB/day throughput with automated lesion segmentation via U-Net architecture (IoU 0.89)

Technical Skills

Certifications: AWS Machine Learning Specialty, AWS Solutions Architect Professional

Analytics: Python (Pandas/NumPy/Scikit-learn), R, SQL, MATLAB, Statistical Modeling, Time Series Analysis

ML/AI: TensorFlow, PyTorch, XGBoost, CuDA, Deep Learning, Natural language processing

Visualization: Tableau, Power BI, Matplotlib/Seaborn, D3.js, Plotly, LookerML, RapidsML

Cloud/Data: AWS (S3/Redshift/SageMaker), Snowflake, PostgreSQL, Apache Spark, Kafka, ETL Pipelines